# Chengyu Liu

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Research Interests: Generative Modeling / Machine Unlearning / Embodied AI and VLA

## Education

**National University of Singapore (NUS)** Aug. 2025 – Present

M.Sc. in Electrical Engineering

- **Cumulative GPA: 4.44/5.0**
- Selected coursework: Adaptive Control Systems; Differentiable and Probabilistic Computing; Visual Computing

**Beijing Institute of Technology (BIT)** Sep. 2021 – Jun. 2025

B.Eng. in Intelligent Unmanned Systems, Xu Teli Honors College

- Outstanding Graduate; average score: 89.4/100; top 12%
- Exchange programs: RWTH Aachen University (Aug. 2023; credit-bearing coursework in Robot Kinematics, SLAM, and Automation Simulation; German grade: 2.3, "good," 1.0 highest) and NUS Summer Data Science Program (Jul. 2023)

## Publications

**Semantic Surgery: Zero-Shot Concept Erasure in Diffusion Models**

*Lexiang Xiong,* ***Chengyu Liu****, Jingwen Ye, Yan Liu, Yuecong Xu* **NeurIPS 2025** [Paper] [Code]

## Conference Papers

**Composite Learning Adaptive Impedance Control of Robot Manipulators under Insufficient Excitation**

*Yang Ding, Yuxiao Liu, Jiajun Shen,* ***Chengyu Liu****, Xiaolong Yu, Wei Wang*

SYS-ELE 2026 Conference Paper; Outstanding Paper, 2026

## Research Experience

**Smooth Flow-Matching Action Policies for Robot Manipulation**

Beihang University, Research Intern; Supervisor: Prof. Jianwei Niu Jun. 2026 – Present

- Investigating chunk-boundary discontinuities and hesitation in rolling execution of flow-matching action policies, focusing on closed-loop trajectory smoothness and robustness under distribution shift
- Built an ACT training-to-rollout baseline with trajectory metrics including chunk-overlap RMSE, NSPARC, NLDLJ, action delta, and jerk
- Designing shift-aware regulation of A2A-informed sources in LIBERO and LIBERO-Long, with Franka validation planned after simulation gates are met

**Systematic Robustness Audit of Graph Unlearning** Sep. 2025 – Present

Working Paper

- Investigating graph-unlearning robustness under adversarially selected deletion requests and heterogeneous graph, dataset, and request settings
- Designing a systematic audit across six graph-unlearning methods and vulnerability fingerprints based on retrain gap, prediction shift, hop-distance decay, and update-detection behavior

### Concept Erasure in Diffusion Models — Sep. 2024 – Jun. 2025

NUS (Suzhou) Research Institute & Beijing Institute of Technology, Undergraduate Thesis

- Developed a zero-shot concept-erasure framework based on text-embedding intervention, enabling precise removal of specified semantics from pretrained text-to-image diffusion models without retraining
- Benchmarked concept-unlearning methods including linear weight updates, distribution matching, negative-guidance imitation, and adapter suppression; evaluated erasure efficacy, robustness, and generation quality
- Built the end-to-end experimental pipeline with PyTorch and Stable Diffusion; used CLIP Score and FID to quantify concept suppression and generation-quality preservation

## Internship Experience

### Aerospace Information Research Institute, Chinese Academy of Sciences

Research Intern — Jul. 2024 – Aug. 2024

- Reviewed research on object detection in synthetic-aperture radar (SAR) imagery and contributed to a ship-detection project
- Trained and evaluated a two-stage SAR ship detector using the detectron2 framework
- Built data-preprocessing and experiment-analysis workflows to support model validation and result reporting

## Selected Engineering Project

### Xiao Yu: Health-Service RAG Assistant (Jiaqi Health)

- Built an end-to-end domain-specific RAG system using Langchain-Chatchat, FAISS, Qwen embeddings, and Qwen LLMs for health-service knowledge retrieval and grounded question answering
- Designed a knowledge-ingestion pipeline from source documents to verified JSON and FAISS indexes, retaining prices, eligibility, safety constraints, and membership rules
- Delivered Next.js and Taro WeChat clients through a shared streaming API; aligned request payloads across both clients and automated tests
- Added configurable retrieval thresholds, source evidence, and tests for knowledge-base behavior and health-advice boundaries

## Technical Skills

- **Programming:** Python, C/C++, MATLAB, JavaScript/TypeScript
- **Machine Learning:** PyTorch, NumPy, Stable Diffusion, CLIP, detectron2, Langchain-Chatchat, FAISS
- **Engineering:** Linux, Git, Next.js, Taro, OpenAI-compatible APIs, model training, and experimental evaluation

## Honors & Awards

- Outstanding Graduate, Beijing Institute of Technology (2025)
- Outstanding Paper, SYS-ELE 2026 — The 2nd Conference on Systems Engineering and Electronics
- Honorable Mention, Mathematical Contest in Modeling (MCM)
- University scholarships, awarded multiple times; approximately RMB 50,000 in total